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Sudden death due to a novel nonsense mutation in Marfan syndrome

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A R T I C L E I N F O

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A B S T R A C T

Background: Marfan syndrome is a hereditary connective tissue disease accompanied by autosomal dominant inheritance; that mainly arises from a mutation in the fibrillin-1 gene (FBN1). Aortic dissection and rupture are the common and lethal complications of MFS and may cause sudden unexpected death.
Method: A man aged 34 was admitted to the hospital due to persistent pain in his abdomen 12 h post-drinking and suddenly died 10 h later. A forensic autopsy was performed to identify the underlying mechanism of death. Due to the high suspected of MFS, Sanger sequencing was performed, and a novel mutation was detected in the deceased. To clarify the underlying mechanism of this mutation, real-time quantitative polymerase chain reaction was conducted and Western blot analysis was performed in vitro.
Results: A novel PTC mutation c.933C > A in FBN1 was found. Through family history inspection and Sanger sequencing, other MFS patients in the present family were confirmed. The pathologic changes in the aorta in the present case showed media cystic degeneration, disordered arrangement of elastic fibers and a significant reduction in fibrillin 1 compared with the control. The mutation led to significant reduction inFBN1 mRNA and fibrillin-1 in cells in vitro, and overexpression of phospho-Smad2 was observed.
Conclusion: We confirmed a novel pathogenic PTC mutation in the FBN1gene through Sanger sequencing, and the pathological changes and underlying mechanisms were also identified. The present work not only extends the pathogenic mutation spectrum of MFS, but also stresses the role of forensic autopsy, genetic analysis and functional validation of novel mutations in cases of sudden death associated with congenital diseases.

1. Introduction

Marfan syndrome (MFS) is a hereditary connective tissue disease with autosomal dominant inheritance; that mainly arises from the mutation in fibrillin-1 gene (FBN1) located on chromosome 15. The morbidity rate of MFS is estimated to be 1/5,000, with no sex biases [1,2]. Individual variation is common in the clinical manifestations of MFS. It has been reported that the clinical spectrum of MFS includes more than 30 different medical characteristics [3]. Aortic dissection and rupture are the most important and life-threatening manifestations of MFS. In the United States, aortic dissections cause 10,000 deaths and are associated with more than 17,000 deaths annually [4,5]. If the patients with MFS are treated in time, their life span can be greatly extended [6].

In 1931, Weve discovered that MFS was heredity and was mainly caused by FBN1 gene mutations and rarely caused by mutations in TGFBR2 or TGFBR1[3,7]. It has been reported that the incidence of detection of mutations in the FBN1 gene may be no less than 90% [8,9]. As of 2014, the FBN1 mutation database (http://www.umd.be/ FBN1/) had reported more than 3000 mutations, most of which were unique, and only 12% were recurrent [10]. The number of known mutations may be false, because some identified FBN1 mutations may not be entered into the database [11].

Missense mutations, premature stop codons and splicing mutations are the main types of FBN1 gene mutations [12]. Missense mutations ranked first, and were found in approximately two-thirds of FBN1 mutation cases. Insertions, deletions, or replications account for approximately 10% to 15% of mutations, most of which produce premature stop codons (PTCs). Splicing mutations account for 10–15%, and most of them alter the canonical splicing sequence at the exon/intron boundary. Other mutations include small in-box deletions and multiple exon deletions, while the absence of FBN1 gene rarely occurrs [13].

The FBN1 gene encodes fibrillin-1, an extracellular matrix macromolecule that can be assembled into microfibrils, which are critical in the extracellular matrix. Most PTC mutations can produce less FBN1

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mRNA through nonsense-mediated mRNA decay (NMD), thus leading to a reduction in the production of the normal fibrillin-1 protein [12,14–16]. FBN1 mutation may affect the deposition of elastic fibers and the regulation of cytokines by regulating the TGF-β signaling pathway, thereby promoting the production of aortic lesions [17].

Here, we present a young man diagnosed with MFS through forensic autopsy and genetic testing. A novel nonsense mutation (c.933C > A (p. Tyr311*)) in FBN1 was detected in his family. This novel mutation may affect the level of FBN1 mRNA through NMD, an overactive TGF-β signaling pathway, resulting in aortic dissection.

2. Material and methods

2.1. Clinical summary of patient

A man aged 34 was admitted to the hospital due to persistent pain in his abdomen post-drinking 12 h prior. The pain was worse in the upper right abdomen, and it did not radiate to other parts. The pain was characterized by tenderness and rebound, and Murphy's test was positive. CT scan showed little pericardial effusion. Due to the high suspicion of cholecystitis, gastric acid suppression was applied and the pain was relieved. However, 9 h post-administration, the patient's condition suddenly deteriorated and he died 1 h later.

Due to the suspicion of medical malpractice in the direct relatives of the deceased, forensic pathologists were asked to determine the real cause of death.

2.2. Genetic analysis

We derived genomic DNA from the deceased's blood, prepared a library by interrupting the DNA, and then applied BGI V4 chip to capture and enrich the DNA of target exons and nearby shearing zones, and finally used the MGISEQ-2000 sequencing platform to detect mutations. The quality control index of sequencing data was as follows: the average sequencing depth of the target area was ≥ 180X, and the proportion of sites with an average depth of the target area >20X was >95%.

We balanced the difference between sequenced fragments with the UCSC hg19 human reference genome by BWA to remove duplications, GATK for base quality correction SNV, INDEL and genotype detection, and Exome Depth to detect copy number variation at the exon level.

The pathopoiesis classification of variants is built on the American Society for Medical Genetics and Genomics (ACMG) and the American Society for Molecular Pathology (AMP) Sequence Variation Interpretation Guidelines, and refers to the Clin Gen Sequence Variation Interpretation Working Group and the British Society for Clinical Genomic Sciences (ACGS).

2.3. Extension of the investigation to the family of the deceased

In view of the high susceptibility of the lineal relatives of MFS patients, we collected related information in this case and acquired their agreements. The deceased's father and aunt and the aunt's younger son showed wrist and thumb signs. The deceased's uncle died at age42, and the cause of death was suggested to be sudden cardiac death even without autopsy. The deceased had 2 sons, aged 10 years and 6 months, and wrist and thumb signs appeared on the younger son. Related information is collected in Table 1. Genetic testing was recommended for the large family and their agreements were acquired, however, due to the COVID-19 outbreak, only some of them were tested.

2.4. Prediction of this novel mutation

To identify the underlying pathogenicity of this novel mutation, we input it into Mutationtaster(http://www.mutationtaster.org/cgi-bin/MutationTaster/MutationTaster69.cgi).

2.5. Functional validation of the mutation

2.5.1. Construction of expression plasmids

A pEGFP-FBN1 plasmid that expressed full-length FBN1 (GenBank: NM_000138.4) cDNA was supplied by Wu Nan (Medical Research Center of Orthopedics, Chinese Academy of Medical Sciences, Beijing, China) and Li Xiaoxin (Medical Research Center & Department of Central Laboratory, Peking Union Medical College Hospital, Peking Union Medical College and Chinese Academy of Medical Sciences, Beijing, China). Homologous recombination was used to design mutant primers (Table2). The FBN1 wild-type plasmid was amplified with a Toyobo KOD-Plus-Neo Amplification Kit (KOD 401), and the PCR products were collected using a 1% agarose gel. The reconstituted reaction solution included 1 μl PCR products, 2 μl Exnase II, 4 μl 5 X CE II Buffer and 13 μl ddH2O. After mixing, the mixture was reacted at 37 °C for 30 min, and immediately placed in an ice bath for 5 min. Then, 100 ul of ToP10 competent cells was added to the reconstituted products, mixed well at 42 °C for 60 s, placed in an ice water bath for 120 s, and spread evenly on a kanamycin-resistant LB plate at 37 °C overnight. A single clone was picked for PCR identification and Sanger sequencing was applied to ensure that the FBN1-C933A mutant plasmid was successfully built.

2.6. Cell culture and transfection

We cultured HEK293T cells with 10% FBS and 90% DMEM in a CO2 incubator at 37 °C and 5% carbon dioxide. We inoculated the cells in six-well plates, and transfected them with 7.5 μl Lipofectamine™ 3000 Transfection Reagent each plate. Six hours after the transfection, fresh DMEM was used to replace the culture medium, and 48 h later, we collected cells for the next experiment.

Table 2
Primer Sequences of PCR Products of cDNA Amplification.

Amplicon	Forward Primer 5′→3′	Reverse Primer 5′→3′
β-actin	AAgAgCTACgAgCTgCCTgA	TggATgCCACAggACTCCAT
FBN1	CCAgCTgTgAAgACgTggAC	TggTAgTgCTggAggTAgCC
Smad2	CCCAgCAggAATTgAgCCACA	CTgCTggAgAgCCTgTgTCC

Table 1
Clinical information of the affected individuals in present family.

Characteristic	Proband (III:2)	Proband's younger son (IV:2)	Proband's father (II:2)	Proband's aunt (II:4)	Proband's cousin (III:5)
Gender	M	M	M	F	M
Wrist and thumb sign	+	+	+	+	+
Pectus Excavatum	+	−	+	−	+
Plain Flat Foot	+	+	+	+	+
Facial features	+	+	+	+	+
Myopia	−	−	−	−	−
Retinal detachment	Unknown	−	−	−	−
Overgrowth of the long bones	+	−	+	+	+
Aortic dissection	+	Unknown	Unknown	Unknown	Unknown

2.7. Real-time quantitative polymerase chain reaction

Total RNA of the cultured cells was prepared according to the manufacturer's instructions, and cDNA was synthesized according to the instructions for the HiScript III RT SuperMix for qPCR (+gDNA wiper) (VAZYME R323-01). RT-qPCR was conducted following the instruction of SYBR Green Master Mix (VAZYME Q311-02) introduction. The primer sequences are listed in Table2.

2.8. Western blot analysis

RIPA lysis buffer (Servicebio G2033), PMSF (Servicebio G2008), 50 × cocktail protease inhibitor (Servicebio G2006), and phosphorylated protease inhibitor (Servicebio G2007) were added to the cell plates for lysis, and then an ultrasonic cell disintegrator was used to completely lyse the cells. We used a BCA kit (Servicebio G2026) to calculate the protein concentration and added 5 × protein loading buffer (non-reduced type) (Servicebio G2030). An 8% SDS polyacrylamide gel was used to separate the protein samples, and nitrocellulose membranes were used to blot the proteins. BSA (5%)was used to block the membranes for 1 h (RT), and the membranes were incubated for 14 h at 4 ℃ in 5% BSA which included primary antibodies (fibrillin-1 antibody, Affinity Biosciences, AF0429, 1:500; phospho-Smad2 (Ser245/250/255) antibody, Cell Signaling Technology, #3104, 1:1000; anti -GAPDH Rabbit pAb, Servicebio GB11002, 1:2000). The membranes were washed 3 times and incubated for 1 h at room temperature with DyLight 800, and goat anti rabbit IgG (Immunoway, RS23920, 1:15000). Then the membranes were visualized in ODYSSEY CLx (LI-COR CLX-1627) and the bands were quantified in ImageJ (Version 1.47 2013). The experiments were repeated 3 times with different Cell samples.

2.9. Statistical analyses

SPSS Statistics 22.0 (SPSS Inc.,) was applied to analyze the experimental data, and the results are displayed as the mean ± S.E.M. The statistical method used in the present study was ANOVA. P less than 0.05 was regarded as statistically significant. GraphPad Prism software was used to construct the related graphs.

3. Results

3.1. Autopsy findings

The decedent was 195 cm, and the length of his arms and legs were 81 cm and 96 cm respectively. His fingers were also long and thin with the joints enlarged, spidery and his metacarpophalangeal joint of the right big toe dislocated. His chest also had a caved-in look, and his spine showed scoliosis. The pericardial cavity was filled with blood and blotted, weighting approximately 700 g. Upon opening the aorta, a 13.0 cm long dissection from the root of the aorta to the descending aorta was found. The entrance of the dissection caused horizontal ruptures of the internal layer of ascending and descending aorta sized 2.0 cm, 1.5 cm respectively. The blood found its way through the aorta into the pericardial, causing a horizontal rupture of the membrane at the root of aorta sized 0.8 cm.

To better understand the condition of aortic lesions, HE, Alcian blue, EVG, and Masson staining were performed on the present case and a traumatic aorta dissection control. Cystic mucosal degeneration was observed by HE staining and confirmed by Alcian blue staining (Fig. 1A, B, G, H). The elastic fibers varied in length and were arranged in a disorderly manner by EVG staining and Masson staining (Fig. 1 C, D, I, J).

Fibrillin-1 immunohistochemical staining was performed according to the instructions of the primary antibody (Affinity Biosciences, AF0429, 1:200) on the present case and control to identify the role of the mutation on fibrillin-1 in aortic lesions. The expression of fibrillin-1 in the lesions was significantly reduced compared with that in the control (Fig. 1 E, F, K, L).

The cause of death was confirmed by the forensic autopsy, and the underlying disease was confirmed to be MFS. In view of the high susceptibility of the lineal relatives of MFS patients, we recommend genetic testing of the present patient and acquired consent.

3.2. Genetic analysis results

In extron 9 of the FBN1 gene in present patient, we detected a nonsense mutation, c.933C > A (p. Tyr311*), in present patient. The mutation was novel and heterozygous. It was absent from numerous population databases, such as the 1000 Genomes Project, Clin Var, Exome Aggregation Consortium, dbSNP, andgnomAD. According to the suggestions of the American College of Medical Genetics (ACMG), this mutation was considered to be pathogenic and was classified as PVS1, PM2, and PP4.

3.3. Extension of the investigation to the big family

Unfortunately, the novel mutation c.933C > A was present in his father(II 2), aunt(II 4), younger son (IV2), and absent in his sister, his sister's children, his wife and his older son. Due to the outbreak of COVID-19, his aunt' s younger son was abroad and we recommended genetic testing for his positive physical signs, at present, the result was still unknown (Fig. 2).

Routine physical examination yearly was recommended to the mutation carriers in present family to record the onset of the positive signs. According to the clinical guideline of MFS in China the cardiovascular complications was stressed as soon as they arrived 18 years old.

3.4. Prediction of this novel mutation

Mutations in Mutationtaster showed disease causing results. The summary of the analysis results can be concluded as NMD, amino acid sequence altered, splice junction changes and possible effect on protein features.

3.5. Functional validation of the mutation

The FBN1 mRNA content and expression of fibrillin-1 were reduced significantly in mutated cells in vitro compared with the control. Overexpression of phospho-Smad2 (Ser245/250/255) was also observed, which indicated overactivity of the TGF-β signal pathway (Fig. 3).

3.6. Genetic counselings to the big family

In 2018, we performed a forensic autopsy on a 30-year-old female who was finally diagnosed as Marfan Syndrome through forensic autopsy and genetic analysis. Considering the risk of the first-degree relatives of the deceased patient affected by MFS and at a risk of TAAD, we investigated the family thoroughly, with some positive signs and the same mutation in the deceased's little daughter. And since then we made a collaboration with some cardiologist performed physical examination annually to recorded the onset of the positive signs of the little daughter to better understand the underlying mechanisms of MFS.

The positive physical signs in some members in present big family arouse our attention, and the cardiologist suggested genetic testing with the same mutation detected. Echocardiographic examination was recommended for the affected members right time and 6 months later to determine the diameter of the aortic root and ascending aorta and the rate of increase. The frequency of monitoring thereafter is determined by the diameter and rate of growth of the aorta. An annual eye assessment was also recommended.

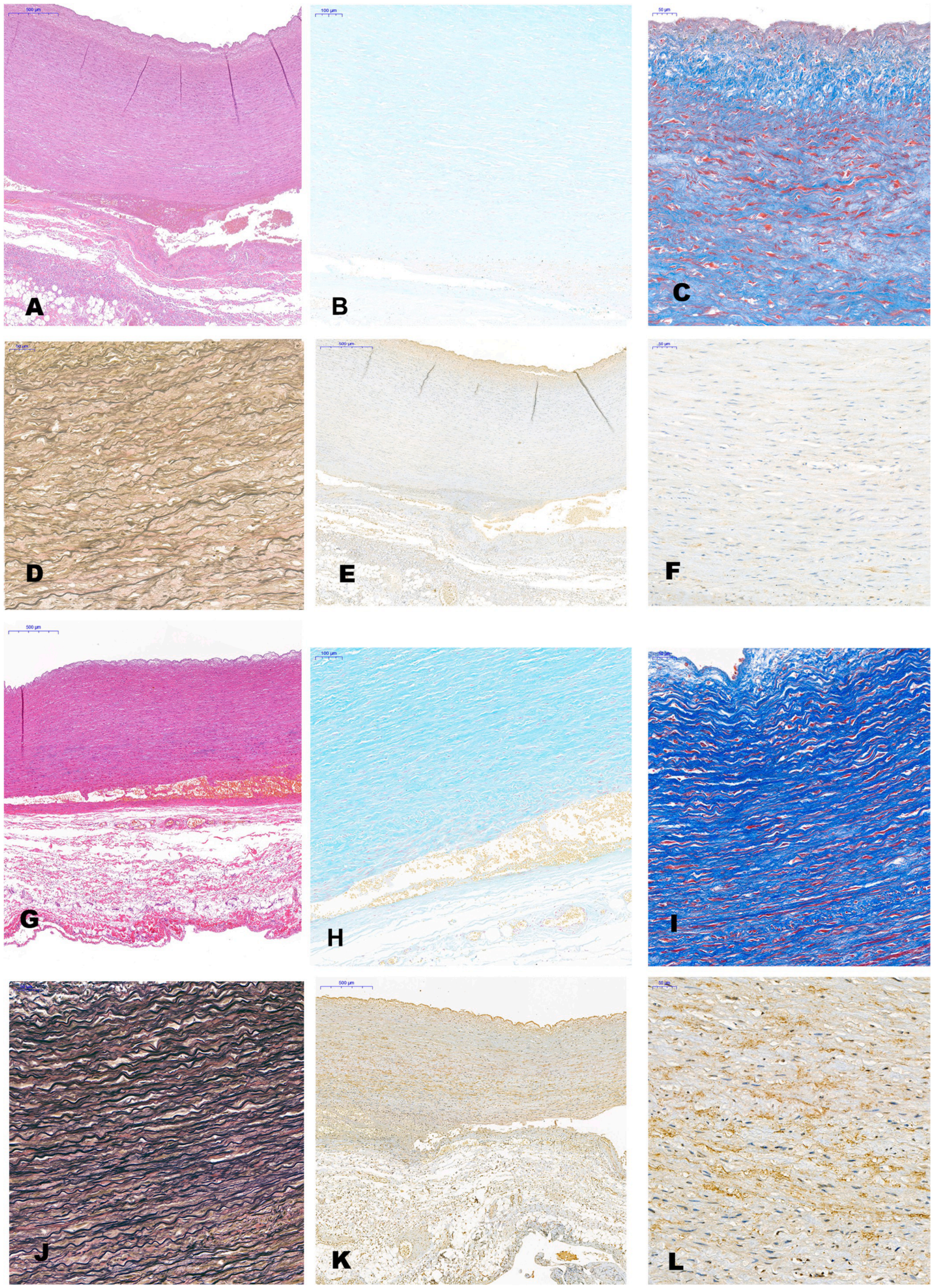

Fig. 1. The pathologic changes in the aorta in the present case, which showed media cystic degeneration, disordered arrangement of elastic fibers and a significant reduction in fibrillin 1 compared with traumatic aortic rupture in a 40-year-old. A-F, present case, G-H, traumatic aortic rupture control; A, G HE 40×; B, H Alcian blue staining 100×; C, I Masson 200×; D, J EVG staining 200×; E, K fibrillin-1 immunohistochemical staining 40×; F, L fibrillin-1 immunohistochemical staining 200×. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

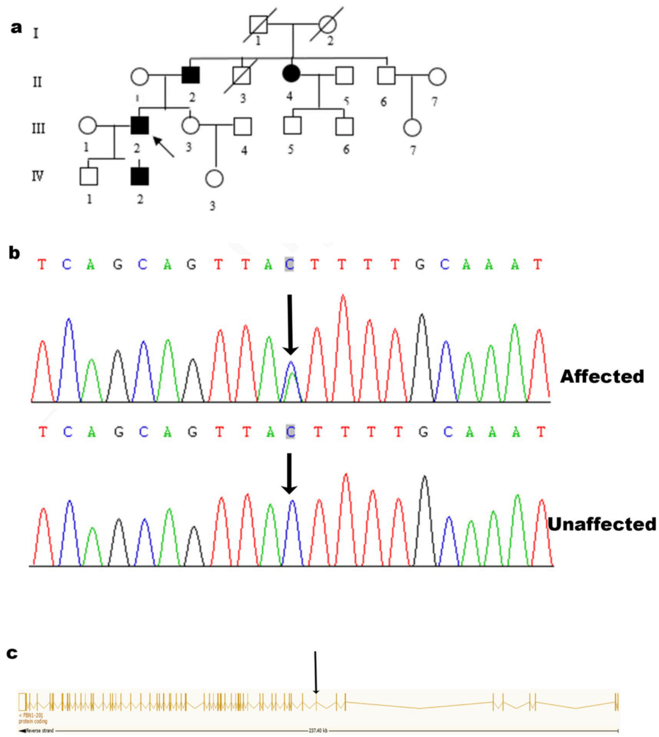

Fig. 2. a, Pedigree showing three affected members. b, Chromatogram of FBN1 heterozygous premature stop codons mutation. c, Genomic organization of FBN1 and the mutation in exon 9.

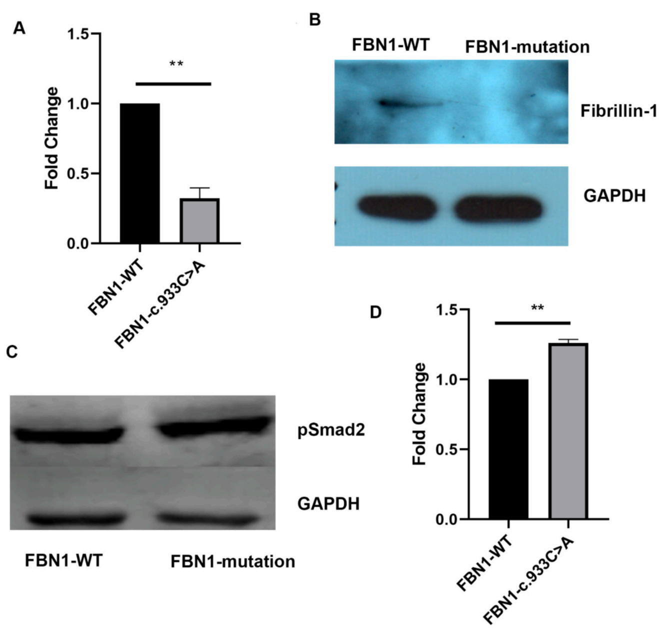

Fig. 3. A, The relative amount of FBN1 mRNA detected in Vitro indicated that the mutation significantly affected the process; B, western blot analysis showed that fibrillin-1 was absent in the mutated cells, C,D, Overexpression of phospho-Smad2 (Ser245/250/255) in mutated cells.

4. Discussion

In the present research, a novel PTC mutation c.933C > A (p. Tyr311*) in the FBN1 gene was detected in a deceased individual whose death was confirmed to be aortic dissection rupture. Through family history inspection and Sanger sequencing, other MFS patients in the present family were confirmed. The function of the mutation was verified in vitro cell experiments.

Many studies stress the relationship between the genotype and phenotype of MFS, and it has been found that the genotype and phenotype of MFS are closely related [18–25]. The location and type of gene mutations are critical for severe phenotypes. Although known FBN1 mutations are spread throughout the gene, the specific region of exons 24–32 is the main region that causes neonatal MFS and severely atypical MFS phenotypes [23]. Mutations in exons 24–32 can predict severe cardiovascular disease and are significantly related to the patient's prognosis.

It was reported that frameshift mutations and nonsense mutations preferentially induce aortic dissection in MFS compared with other mutations in a study focused on the interrelation between the genotype and phenotype of Chinese MFS patients [18]. Compared with missense mutations, patients with PTC or spliced mutations have aortic events at younger ages. MFS patients with PTC mutations have obvious aortic disease and skeletal features, whereas the risk of severe ocular manifestations is much lower [26]. The relationship between mutations in other regions and severe cardiovascular disease was still unclear, previous studies indicated that even the mutations in intron can cause poor prognosis, the mutation and mutation type in the present family together contributed to the positive physical signs [27].

It is well known that premature termination of translation can lead to a decrease in mRNA levels in the organism [16]. In eukaryotes, this incidence is called nonsense-mediated mRNA decay (NMD) and is activated by PTC in the context of atypical sequences. It has been reported that NMD can affect as large as 30% of the expression of the genome [28]. The proteins produced by pathogenic mutant genes are defective and may damage the organism. Due to the existence of NMD, PTC rarely produces truncated proteins. Previous studies have shown that most PTC mutations played a pivotal part in the pathogenic mechanism of MFS, and NMD is not the only mechanism in PTC mutations [29]. The relative contents of FBN1 mRNA vary in organs and are much higher in blood than in affected fibroblasts, whereas no mutant alleles were detected in fibroblasts without NMD inhibition, suggesting that tissue-specific degradation of mRNA also makes sense [30,31]. Most premature stop codon mutations will result in a reduction in the production of the normally fibrillin-1 protein, which was also confirmed in the pathological changes of affected aorta in the present case. The FBN1 mRNA content and the expression of fibrillin-1 in the vitro cell experiment was consistent with previous studies.

Previous studies uncovered the main pathogenic mechanisms of thoracic aortic aneurysm and dissection (TAAD): variation of extracellular matrix protein and TGF-β signal transduction protein, smooth muscle cell (SMC) contractile dysfunction and impaired SMC differentiation and proliferation [32]. Fibrillin-1 can regulate the activity of TGF-β by releasing TGF-β from the ECM after receiving various stimuli. Pathogenic mutations of FBN1 could increase the activity of TGF-β and cause the TGF-β signal cascade, resulting in the thinning and rupture of the elastic fibers of the aortic smooth muscle, which eventually causes TAAD. This process is considered to be the main pathogenesis of TAAD caused by MFS [33,34]. It is a common view that the dysregulation of TGF-β signaling can explain various clinical characteristics in MFS [34]. The over activity of TGF-β signal pathway confirmed in vitro caused by the novel PTC mutation in the present study may explain the phenotype in the deceased.

Mutations in FBN1 not only result in typical MFS, but also have a critical impact on the pathogenic mechanism of sporadic AD [35]. Variations in the clinical characteristics of MFS may lead to large phenotype variations in tissue distribution, onset time and severity. Even probands with the same FBN1 mutation may display diffident positive physical signs within the same family. For people suspected of having MFS, systematic and long-term clinical follow-up evaluations could be required to make an authentic diagnosis based on medical evidence. Under certain circumstances, especially the proband was confirmed, gene sequence analysis of the FBN1 in the family can help make early diagnosis to avoid such tragedies through medical intervention.

The relationship between sudden death and aortic dissection is well known and in 2017, genetic analysis were recommend in sudden aortic death cases. The critical role of postmortem genetic analysis in genetic disorder diseases has been confirmed in forensic practice. Due to the inherited characteristics the affected directive relatives of the deceased may suffer the same tragedy and early diagnosis could provide clinical suggestions and clinical intervention. Mutations in forensic practices are always associated with sudden death and may display a more severe phenotype than those in clinical practice.

In forensic practice, postmortem genetic analysis is common in sudden death cases, whereas most of the work were finished with mutations detected, though the extension of the investigation of the family and functional validation of the mutations were absent. Through extension of the investigation of the family, the positive signs of the diseases may be also revealed in the family, and genetic analysis can help identify the mutation carriers in the family and medical advice can be delivered to them, so that they can search for medical intervention in time. Considering the characteristics of forensic practice, the functional validation of these mutations was critical for defining the mechanism of sudden death.

In conclusion, we confirmed a novel pathogenic PTC mutation in the FBN1gene through Sanger sequencing, the pathologic changes and underlying mechanisms were also identified. We also found other positive physical characteristics in the deceased's extended family, performed Sanger sequencing with some positive findings, and provided medical advice for avoiding potential future aortic lesions. The present work not only extends the pathogenic mutation spectrum of MFS, but also stresses the roles of forensic autopsy, genetic analysis and functional validation of novel mutations in sudden deaths associated with congenital diseases.

5. Compliance with Ethical standards

All research protocols included human samples were approved by the Human Ethical Commission at Tongji Medical College, HUST [2017] IEC (S059).

Informed consent

Written informed consent and research study for publishing this scientific report was obtained from the direct relative of the decedent in present study.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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